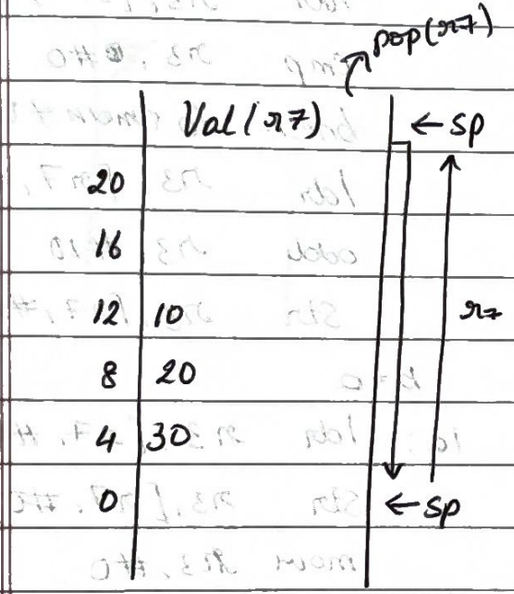
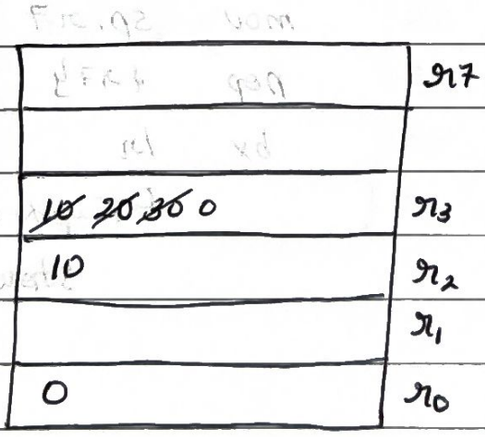
RTOS

1

```
int main (void){
    int a=10, b=20;
    int c = a+b;
}
```

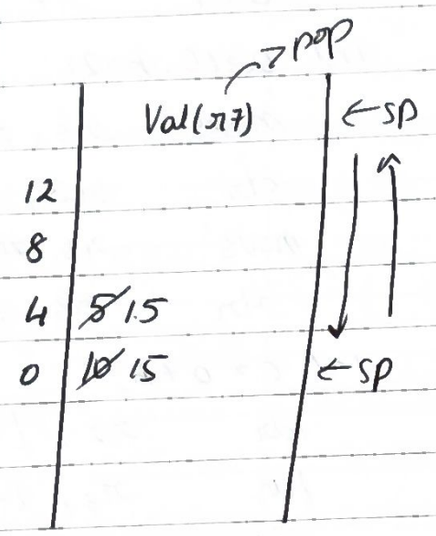
```
00000000 <main>:
#include <stdio.h>
int main (void){
        push    {r7}
        sub     sp, #20
        add     r7, sp, #0
int a=10, b=20;
        movs    r3, #10
        str     r3, [r7, #12]
        movs    r3 #20
        str     r3, [r7, #8]
int c = a+b;
        ldr     r2, [r7, #12]
        ldr     r3, [r7, #8]
        add     r3, r2
        str     r3, [r7, #4]
        movs    r3, #0
        mov     r0, r3
        adds    r7, #20
        mov     sp, r7
        pop     r7
        bx      lr
```

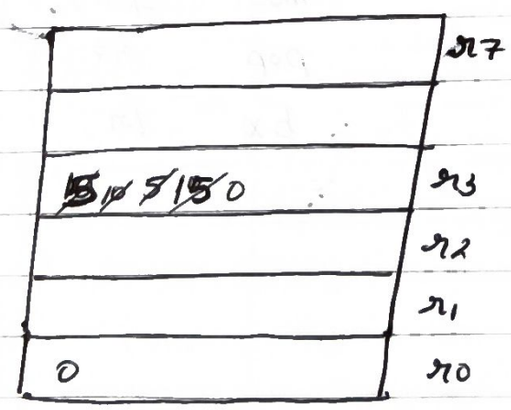




2

```
int main (void){
    int a = 5;
    int b = 10;
    if (a == 0) a = a + 10;
    b = a;
}
```



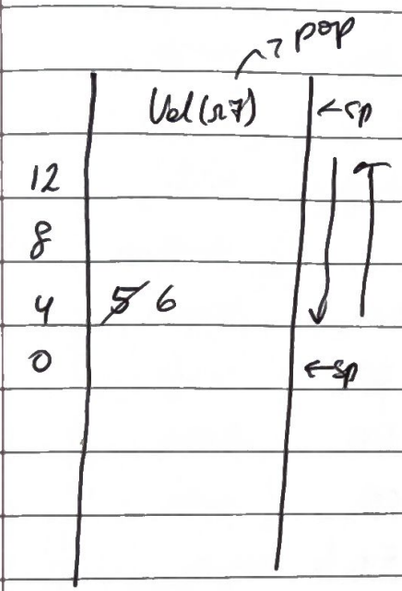


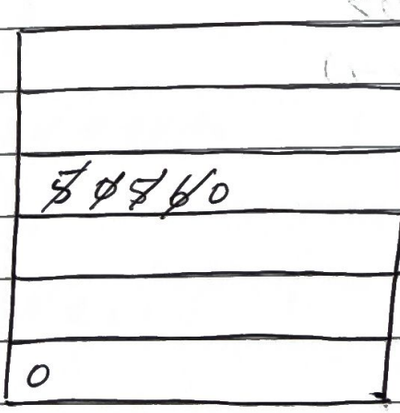

```
00000000 <main>
int main (void){
    push {r7}
    sub  sp, #12
    add  r7, sp, #0
int a = 5
    movs r3, #5
    str  r3, [r7, #4]
int b = 10
    movs r3, #10
    str  r3, [r7, #0]
if (a == 0) a = a + 10;
    ldr  r3, [r7, #4]
    cmp  r3, #0
    bne.n 1a <main + 0x1a>
    ldr  r3, [r7, #4]
    adds r3, #10
    str  r3, [r7, #4]
b = a
1a: ldr  r3, [r7, #4]
    str  r3, [r7, #0]
    movs r3, #0
    mov  r0, r3
    adds r7, #12
    mov  sp, r7
    pop  {r7}
    bx   lr
```

↳ Jump to add
stored in lr

3

```
int main (void){
    int a=5;
    if (a==0)
        a=a+10;
    else
        a=a+1;
    return 0;
}
```






```
00000000 <main>
int main(void){
        push  {r7}
        sub   sp, #12
        add   r7, sp, #0
int a=5
        movs  r3, #5
        str   r3, [r7, #4]
if a==0
        ldr   r3, [r7, #4]
        ~~add~~ r7
        cmp   r3, #0
        bne.n 18 <main+0x18>
    a=a+10;
        ldr   r3, [r7, #4]
        add   r3, #10
        str   r3, [r7, #4]
        b.n   1e <main+0x1e>
    a=a+1;
18:     ldr   r3, [r7, #4]
        add   r3, #1
        str   r3, [r7, #4]
return 0
1e:     movs  r3, #0
        mov   r0, r3
        add   r7, #12
        mov   sp, r7
        pop   {r7}
        bx    lr
```

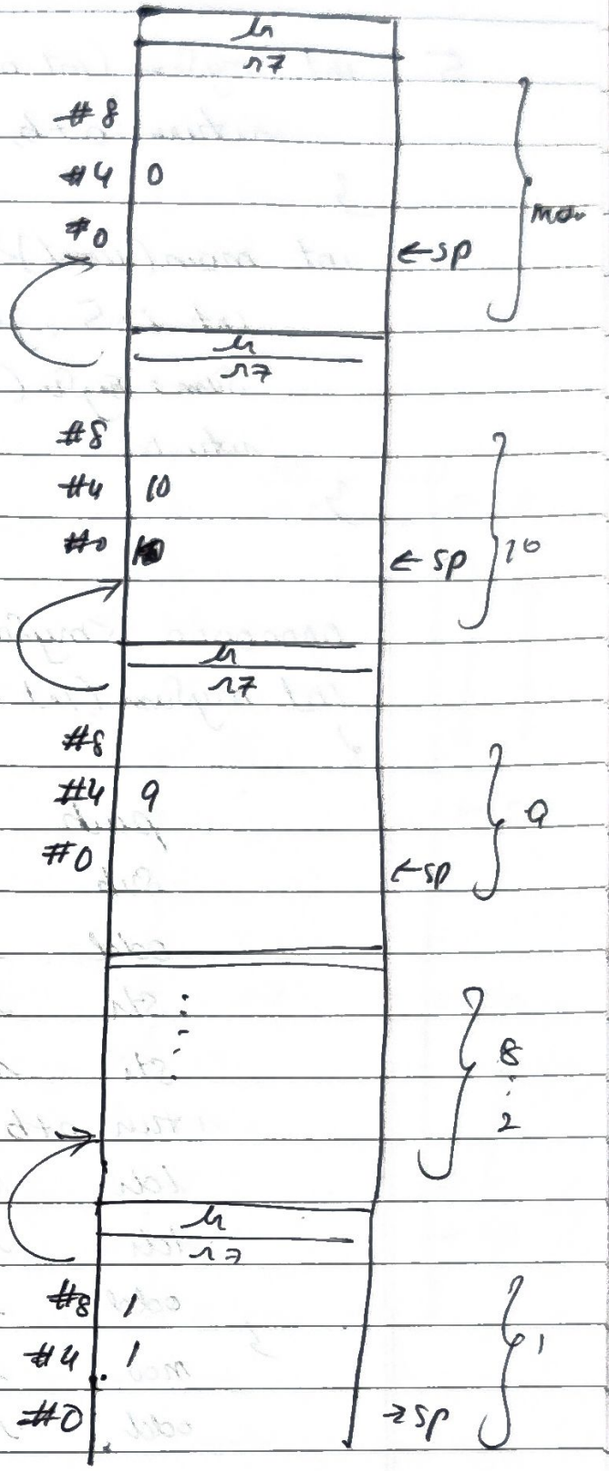
4

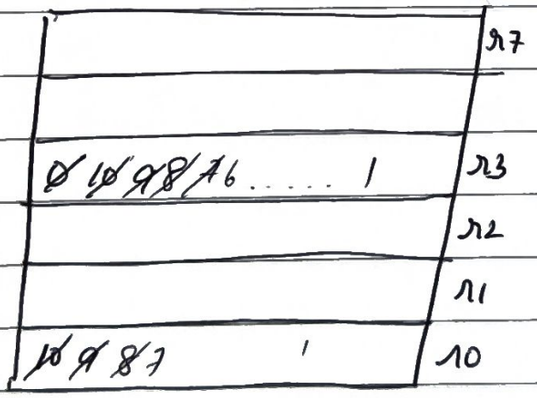
```
int rec_sum (int i){
    if (i<= 1) return 1;
    return 1 + rec_sum (i-1);
}
int main(void){
    int sum=0;
    Sum = rec_sum(10);
    return 0;
}
```

```
00000000 <rec_sum>:
int rec_sum (int i){
0:   push  {r7, lr}
     sub   sp, #8
     add   r7, sp, #0
     str   r0, [r7, #4]
  if (i<= 1) return i
     ldr   r3, [r7, #4]
     cmp   r3, #1
     bgt.n 12 <rec_sum+0x12>
     ldr   r3, [r7, #4]
     b.n   22 <rec_sum+0x22>
  else return 1 + rec_sum (i-1)
12:  ldr   r3, [r7, #4]
     subs  r3, #1
     mov   r0, r3
18:  bl    0 <rec_sum>
     mov r2, r0
     ldr r3 [r7, #4]
     add  r3, r2
}
22:  mov   r0, r3
     adds  r7, #8
     mov   sp, r7
     pop {r7, pc}
```

classmate
Date
Page

```
00000020 <main>
int main(void){
    push {r7, lr}
    sub  sp, #8
    add  r7, sp, #0
  int sum=0
    movs r3, #0
    str  r3, [r7, #4]
  sum = rec_sum(10);
    movs r0, #10
    bl   0 <rec_sum>
    str  r0, [r7, #4]
  return 0;
    movs r3, #0
    mov  r0, r3
    adds r7, #8
    mov  sp, r7
    pop  {r7, pc}
```





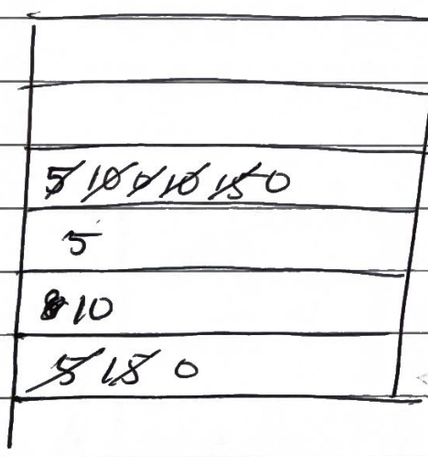
5

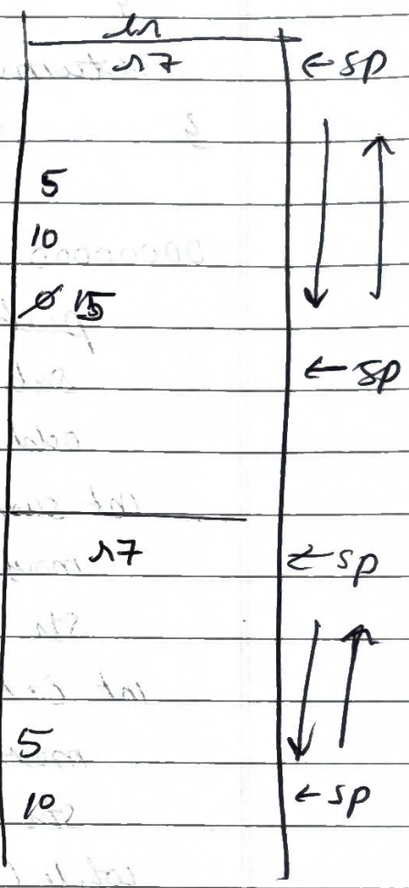
```
int mySum (int a, int b){
    return a+b;
}
int main(void){
    int i=5, j=10, sum=0
    Sum = mySum(i,j);
    return 0;
}
```

```
00000000 <mySum>
int mySum (int a, int b)
{
        push    {r7}
        sub     sp, #12
        add     r7, sp, #0
        str     r0, [r7, #4]
        str     r1, [r7, #0]
    return a+b;
        ldr     r2, [r7, #4]
        ldr     r3, [r7, #0]
        add     r3, r2
    }
        mov     r0, r3
        add     r7, r2
        mov     sp, r7
        pop     {r7}
        bx      lr
```

```
00000000 <main>:
int main (void){
        push    {r7, lr}
        sub     sp, #16
        add     r7, sp, #0
    int i=5, j=10, Sum=0;
```

```
movs   r3, #5
str    r3, [r7, #12]
movs   r3, #10
str    r3, [r7, #8]
movs   r3, #0
str    r3, [r7, #4]
Sum = mysum(i, j);
ldr    r1, [r7, #8]
ldr    r0, [r7, #12]
bl     0 <mySum>
str    r0, [r7, #4]
return 0;
movs   r3, #0
```



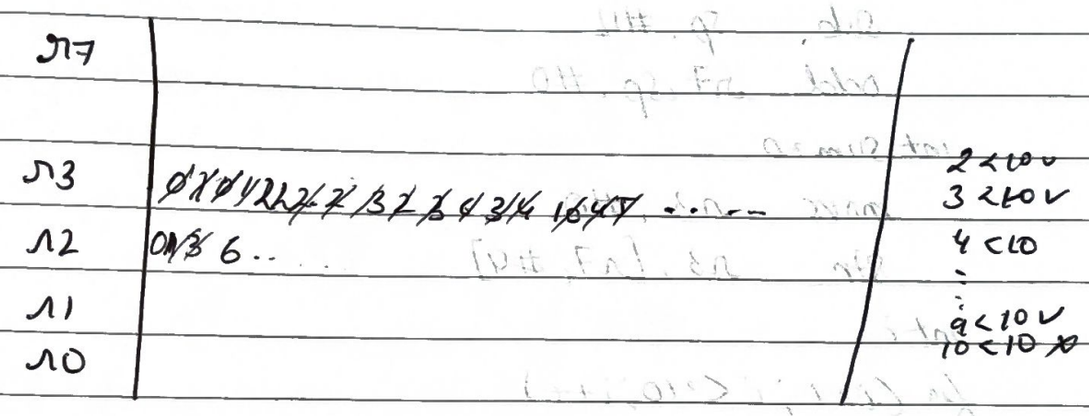


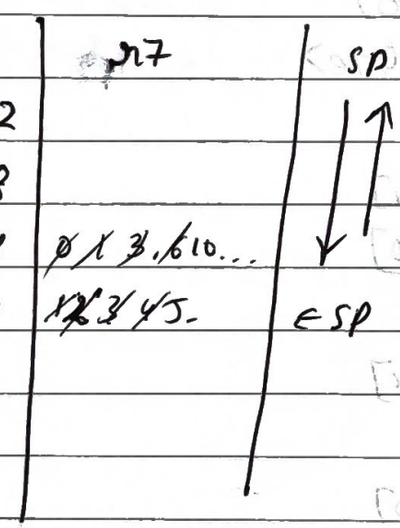
6

```
int main(void)
{
    int sum=0
    int i=1
    while(i<=10){
        sum+=i
        i++;
    }
    return 0;
}
```

```
00000000 <main>
    push {r7}
    sub sp, #12
    add r7, sp, #0
int sum=0
    movs r3, #0
    str r3, [r7, #4]
int i=1
    movs r3, #1
    str r3, [r7, #0]
while(i<=10){
    b.n 1e <main+0x1e>
sum+=i
10: ldr r2, [r7, #4]
    ldr r3, [r7, #0]
    add r3, r2
    str r3, [r7, #4]
i++;
    ldr r3, [r7, #0]
    adds r3, #1
    str r3, [r7, #0]
while(i<=10){
1e: ldr r3, [r7, #0]
```

```
cmp   r3, #10
ble.n 10 <main + 0x10>
}
return 0;
movs  r3, #0
mov   r0, r3
addw  r7, #12
mov   sp, r7
pop   {r7}
bx    lr
```





7

```
int main (void){
    int sum=0;
    int i;
    for (i=1; i<=10; i++)
        sum += i;
    return 0;
}
```

```
00000000 <main>
int main (void){
        push {r7}
        sub  sp, #12
        add  r7, sp, #0
    int sum=0
        movs r3, #0
        str  r3, [r7, #4]
    int i
    for (i=1; i<=10; i++)
        movs r3, #1
        str  r3, [r7, #0]
        b.n  1e <main + 0x1e>
    sum += i
10:     ldr  r2, [r7, #4]
        ldr  r3, [r7, #0]
        add  r3, r2;
        str  r3, [r7, #4]
    for (i=1; i<=10; i++)
        ldr  r3, [r7, #0]
        adds r3, #1
        str  r3, [r7, #0]
1e      ldr  r3, [r7, #0]
        cmp  r3, #10
        bl.n 10 <main + 0x10>
    }
    return 0;
```

```
movs    r3, #0
mov     r0, r3
adds    r7, #12
mov     sp, r7
pop     {r7}
bx      lr
```

